NCTLA · OCTOBER 23, 2026

Generated Is Not Learned

Keeping the thinking on the student's side of the table

Andrew Norris · Instructor of Computer Programming & Development
Fayetteville Technical Community College

YOU LEAVE WITH

- One question to ask of any AI output
- One activity you can run in your center
- Five ways to give AI a better job

ORIENTATION

Andrew Norris

WHO I AM

Instructor of Computer Programming & Development, Fayetteville Technical Community College

Teaching programming at FTCC since 2015
Capstone project leader since 2016
AI instructor since 2023

WHAT I DO

I use all the major AI tools, mostly Claude to write code. In 2026, everything is code.

My tutoring: the Friday Programming Help Lab, one-on-one help for students in the department, about 12 Fridays a semester.

HOW TO USE THIS SESSION

Everything is in the materials; no need to screenshot.

Slides · Blackout Test · Prompts

tinyurl.com/nctla26



MY PERSPECTIVE · VIRTUAL TEACHING ASSISTANT

VITA, 2024

We tried to build it. It was a great way to find out how complex AI-assisted tutoring is.

MEANT TO DO

A virtual teaching assistant for our students.

WHAT WE BUILT

Summer 2024: the chatbot web interface, the logins, and local LLM support.

WHAT WE NEVER SOLVED

“Don’t just give them the answer.”

MY PERSPECTIVE · TODAY

The hard part is the persona

CODING ASSISTANTS

I use AI coding tools every day. They are my working definition of a virtual assistant.

ELIZA, 1966

Weizenbaum's chatbot reflected your words back like a therapist. It did none of the work.

THE TUTORING VERSION

A bot that sounds supportive but leaves the thinking undone.

Help that is real, without an ELIZA situation.

WHAT A PERSON BRINGS

What you have that the bot does not

YOU CAN SEE IT LAND

The pause, the frown, the lean-in, the “oh.” The bot sees only what the student types.

YOU HEAR WHAT WAS NOT ASKED

An answer service waits for a question. A tutor hears the question that was not asked.

You cannot look up a word you have never heard.

ONE QUESTION FOR ANY AI OUTPUT

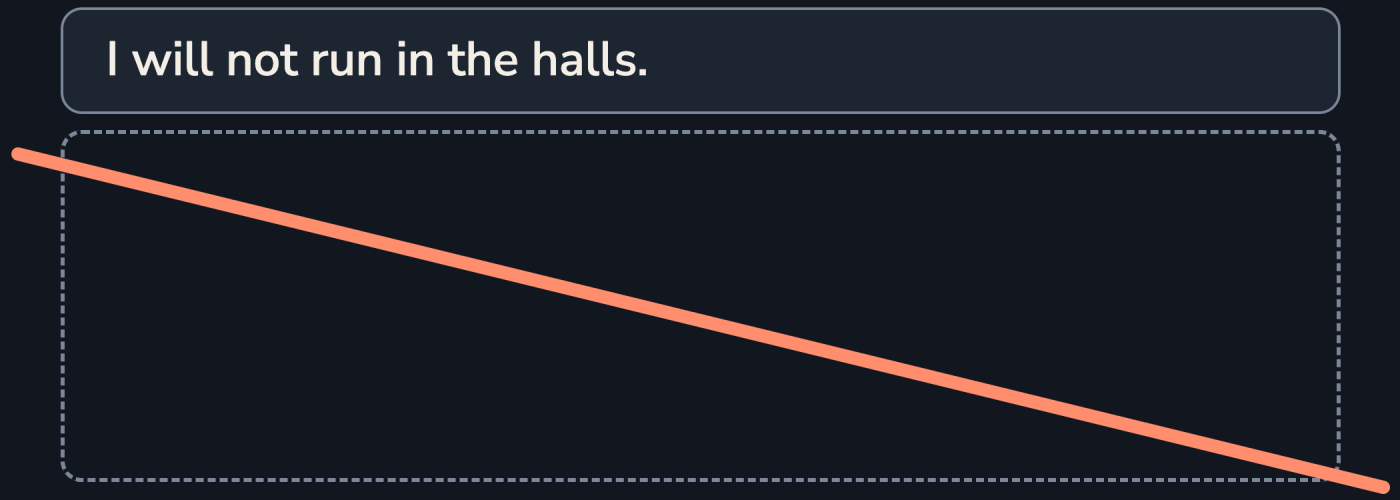
What did the student actually have to think about?

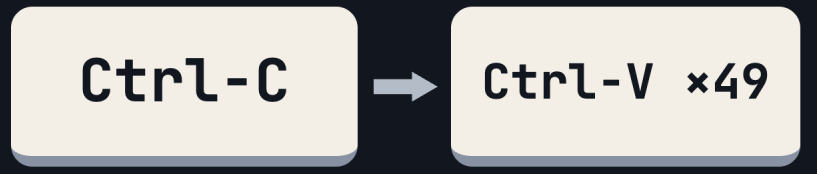
INTO THE CHAT

Think of the last AI-made thing a student showed you. What did they have to think about to get it?

WHEN I WAS IN SCHOOL

50 lines, or Ctrl-C and Ctrl-V





Same page. The page was never the point.

DEPARTMENT OF ACADEMIC OUTPUT · MOTTO: MORE IS BETTER

MORE OUTPUT = MORE LEARNING

“Just do more, faster,
~~and don’t think about it.”~~

DEPARTMENT OF ACADEMIC OUTPUT · MOTTO: MORE IS BETTER

MORE OUTPUT

≠ The thinking part is not optional.

MORE LEARNING

BEFORE THE FIRST CASE · A TUTOR'S OBSERVATION

Two things on the table



Immaculate. Ten pages, every citation formatted.



The copy of the book: the spine was never cracked.

Maybe they read it on Kindle. It's just an observation.

DEPARTMENT OF ACADEMIC OUTPUT · INTAKE WINDOW

You are the reviewers. Stamp each submission in the chat.

```
1 APPROVED
2 RETURNED — no attempt shown
```

Start with the case number.

Feel free to abbreviate to A or R.

A reason is optional: five words or fewer.

Our one metric: MORE OUTPUT = MORE LEARNING

CASE 1 ENG 111

submitted Tuesday 10:15 am

A student brings an outline for a five-page essay on *The Great Gatsby* and the American Dream. It has five themes, three quotes for each theme, and paragraph topics. The student says the outline came from an AI tool.

THE STUDENT SAYS

“I’m ready to write.”

Stamp it: **1 APPROVED** or **1 RETURNED**

CASE 2 MAT 152

submitted Tuesday 1:40 pm

A student is stuck on a statistics problem. The student photographed the worksheet, and AI solved question 4. The solution is correct and explains each formula. The student copied the work.

THE STUDENT SAYS

“That makes sense when I read it.”

Stamp it: **2 APPROVED** or **2 RETURNED**

CASE 3 BIO 168

submitted Wednesday 9:05 pm

A student asked AI for a study guide on the cardiovascular system for tomorrow's exam. The AI made six pages of notes, tables, and mnemonics. The student has reread the guide for an hour.

THE STUDENT SAYS

"I've been over it three times."

Stamp it: **3 APPROVED** or **3 RETURNED**

CASE 4 ENG 111

submitted Thursday 2:40 pm

A student in another section brings an outline for the same Gatsby essay. It has four themes, quotes for each theme, and a topic sentence for each paragraph. The student made it with an AI tool.

THE STUDENT SAYS

“It’s all there. I just need to write it up.”

Stamp it: **4 APPROVED** or **4 RETURNED**

CASE 1 AND CASE 4

Same assignment. Same kind of outline.

CASE 1 · Tuesday

Five themes, three quotes each, paragraph topics.

CASE 4 · Thursday

Four themes, quotes for each, a topic sentence per paragraph.

From the outside, both look ready.

BEHIND THE OUTPUT

CASE 1 · Tuesday

- Wrote a claim first: “Gatsby’s money cannot buy him the past.”
- Asked the AI: “Challenge my evidence. Do not give me quotes.”
- Found the passages that support the claim.
- Rejected two of the AI’s themes, and said why.
- Then asked the AI to format the outline.

CASE 4 · Thursday

- Typed one instruction into the AI: “Do this assignment.”
- skipped
- skipped
- skipped
- The AI produced the outline. Copied it into a document.

Similar output. Different student. Generated is not learned.

THE ONE QUESTION, PART TWO

The cost sits in the checking

GENERATE

Seconds

Fluent, confident, formatted.
Looks finished.

CHECK

Much longer

And checking needs the understanding the shortcut skipped.

The follow-up question: who checks it, and with what?

INTO THE CHAT

You have 60 seconds with the student.

What do you ask?

Type one question that tells case 1 from case 4.

IN A SESSION

Treat AI output as a candidate, not an answer



Candidate accepted. Thinking skipped.

THE TRANSFER CHECK

After the AI, the student does one thing alone

- Explain the idea
- Solve a similar problem
- Find an error
- Defend a choice
- Summarize from memory
- Apply it to a new example

Same skill, new surface, no AI in view.

ONE IDEA WORTH LOOKING INTO

Take the tutor’s words away

Tutor and student work one problem.

blacked out

Student’s words

blacked out

Student’s words

THEN

The student gets a new variant. Only what the student wrote can help.

Who wrote it? That’s who learned it.

BEFORE THE FIVE ROLES

Avoid the Socratic trap

THE TRAP

All questions, no information. For a learner without the words, it is a quiz they cannot pass.

THE FIX

Explain, then ask, then step back. Ask for the stepping back by name.

FIVE AI ROLES THAT LEAVE THE THINKING WITH THE LEARNER

Give the AI one of these jobs

1	**Coach who steps back**	Explains, then asks, then fades its help.
2	**Question generator**	Makes practice and questions, not answers.
3	**Contrast generator**	Gives versions the learner must compare.
4	**Error analyst**	Finds where an attempt first went wrong.
5	**Practice simulator**	Runs an oral review, a scenario or a case.

In every role, the learner makes the decisions and does the explaining.

ROLE 1 OF 5 · COACH WHO STEPS BACK

Guided learning, with the help fading

PRINCIPLE Worked example, then fading: show one, do one with blanks, do one alone.

USE WHEN The learner is new to the topic.

```
# Teach me [topic] step by step
- I am a [level] student in [course]. Stay within [scope].
- I am new to [topic]. Use plain words. Define each new term one time.
- Show one worked example first. Explain each step.
- Then give me a similar problem with some steps blank. Wait for my answer.
- Then give me one problem to do alone. Do not help until I ask.
- When I am stuck, give one hint. Do not give the answer.
- Ask me to explain my reasoning before you tell me if I am correct.
```

PROS

- It explains before it asks, so no Socratic trap.
- It asks for the fading by name.
- It asks for reasoning before it says right or wrong.

CONS

- After a few turns, the tool can drift back to telling.
- The student can still type “just give me the answer.”
- The worked example can be wrong. Check it.

ROLE 2 OF 5 · QUESTION GENERATOR

Vocabulary map

PRINCIPLE Activate what they know. Name what they do not know.

USE WHEN The learner does not know what to ask.

```
# List the terms I do not know yet about [topic]
- I am a [level] student in [course]. Stay within [scope].
- I am a beginner in [topic].
- List 10 terms that an expert uses and a beginner may
not know.
- Give each term a one-sentence definition.
- Mark the 3 terms to learn first.
```

PROS

- Fast and low risk.
- It gives the session an agenda.
- It helps the student who does not know what to ask.

CONS

- A definition can be wrong or too technical.
- A list is not understanding. It can become one more thing to reread.
- Ten terms can swamp a beginner.

ROLE 3 OF 5 · CONTRAST GENERATOR

Two diagrams, explained back

PRINCIPLE Contrasting cases: the learner finds the difference and explains it.

USE WHEN The idea has parts, steps, or flow.

```
# Draw [process or idea] as a diagram
- I am a [level] student in [course]. Stay within [scope].
- Show each part or step as a labeled box.
- Use arrows to show order or cause.
- Then draw a second version with one step wrong.
- Do not explain either diagram.
- Ask me which version is right, and why. Then tell me
which parts of my explanation are wrong.
```

PROS

- Comparing is where the thinking happens.
- The student explains. The tool does not.
- It works for any process or flow.

CONS

- Some tools draw badly. Text diagrams get messy.
- The “right” diagram can be wrong too.
- Check both before the student sees them.

ROLE 4 OF 5 · ERROR ANALYST

Find the error

PRINCIPLE **Learning from errors: find it, show why, repair it.**

USE WHEN The learner can almost do it.

```
# Write a short solution to [problem] that has one error
- I am a [level] student in [course]. Stay within [scope].
- Use an error that beginners often make.
- Do not tell me where the error is.
- When I find it, ask me to show why it is an error.
- When I cannot find it, give one hint at a time.

# Or: Find the first place I went off track
- I am a [level] student in [course]. Stay within [scope].
- Here is my work on [problem].
- Find the first step where my reasoning goes wrong.
- Do not fix it. Ask me one question that helps me find it.
```

PROS

- The output becomes a claim to test.
- It targets the errors beginners make.
- Steps 1 to 4 of the session rule, in one prompt.

CONS

- The tool may plant two errors, or none.
- A weak student can learn the error. End by stating the right method.
- Dosage: label the work as practice, never as a reference.

ROLE 5 OF 5 · PRACTICE SIMULATOR

Teach it back

PRINCIPLE Self-explanation: only the learner's own words count.

USE WHEN The learner thinks they have it.

```
# Check my explanation of [topic]
- I am a [level] student in [course]. Stay within [scope].
- I will explain [topic] in my own words.
- Do not rewrite my explanation.
- Tell me which parts are correct and which parts are not.
- Ask me one question about the weakest part.
```

PROS

- The student does the explaining.
- It shows gaps the student cannot see.
- It works between sessions.

CONS

- The tool may rewrite the explanation anyway.
- It can praise a weak answer. Tell it to be strict.
- The student must type a real explanation. That takes effort.

TAKE WITH YOU

What did the student actually have to think about?

Audit the output

Give the AI a different job

Check transfer with the AI closed

Questions, and questions about my answers

All materials: tinyurl.com/nctla26 · Paper: arxiv.org/abs/2610.06446
Code: github.com/eth-lre/EduardoRL · norrisa@faytechcc.edu · norrisaftcc.github.io